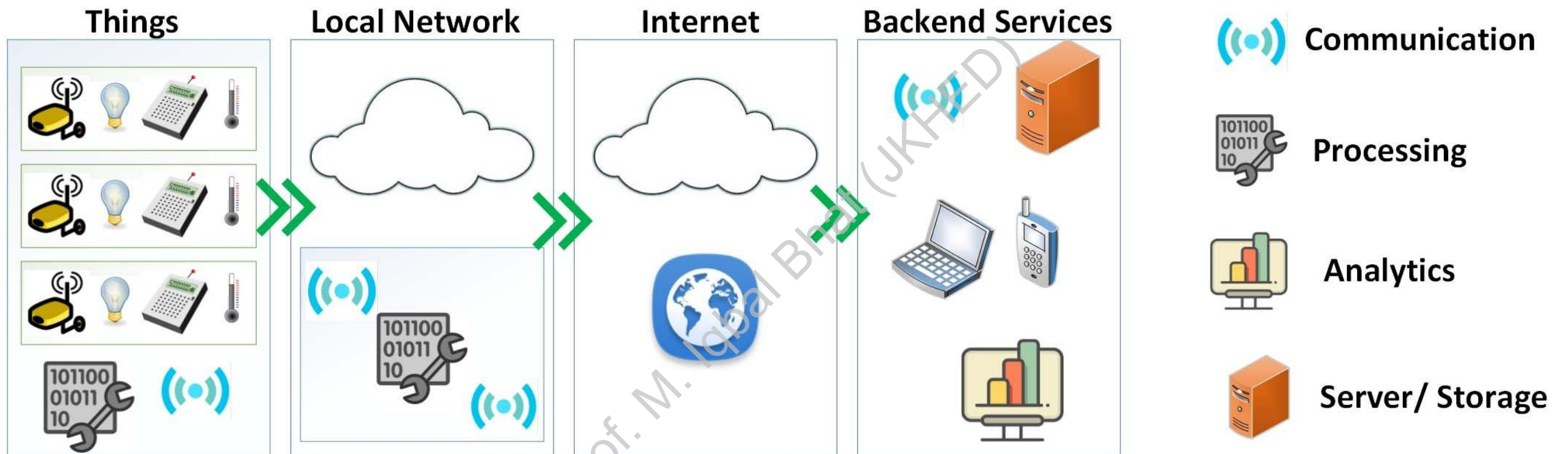
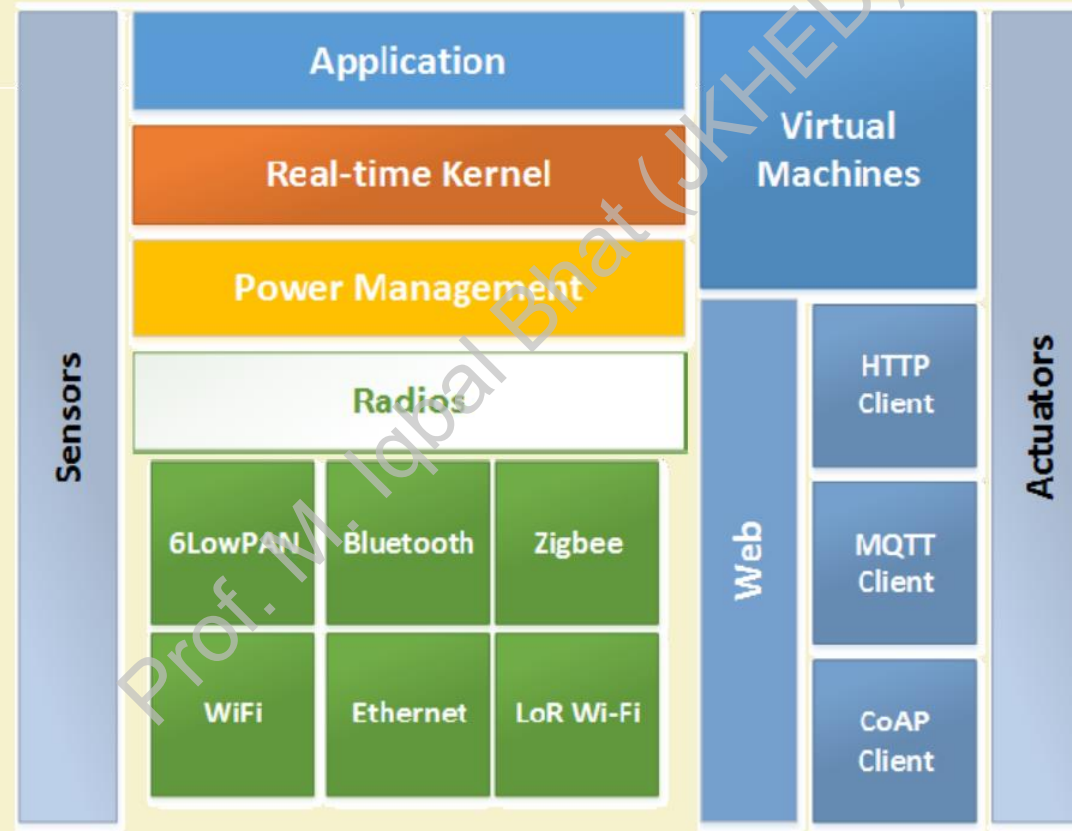


Sensing and Actuation

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IoT Interdependencies



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Sensors

Definition

- ✓ A sensor detects (senses) changes in the ambient conditions or in the state of another device or a system, and forwards or processes this information in a certain manner [1].

“A device which detects or measures a physical property and records, indicates, or otherwise responds to it” [2].

- Oxford Dictionary

References:

1. <http://www.businessdictionary.com/definition/sensor.html>
2. <https://en.oxforddictionaries.com/definition/sensor>

Sensors

- ✓ They perform some input functions by sensing or feeling the physical changes in characteristics of a system in response to a stimuli.
- ✓ For example heat is converted to electrical signals in a temperature sensor, or atmospheric pressure is converted to electrical signals in a barometer.

Transducers

- ✓ Transducers convert or transduce energy of one kind into another.
- ✓ For example, in a sound system, a microphone (input device) converts sound waves into electrical signals for an amplifier to amplify (a process), and a loudspeaker (output device) converts these electrical signals back into sound waves.

Sensor vs. Transducer

- ✓ The word “Transducer” is the collective term used for both **Sensors** which can be used to sense a wide range of different energy forms such as movement, electrical signals, radiant energy, thermal or magnetic energy etc., and **Actuators** which can be used to switch voltages or currents [1].

References:

1. http://www.electronics-tutorials.ws/io/io_1.html

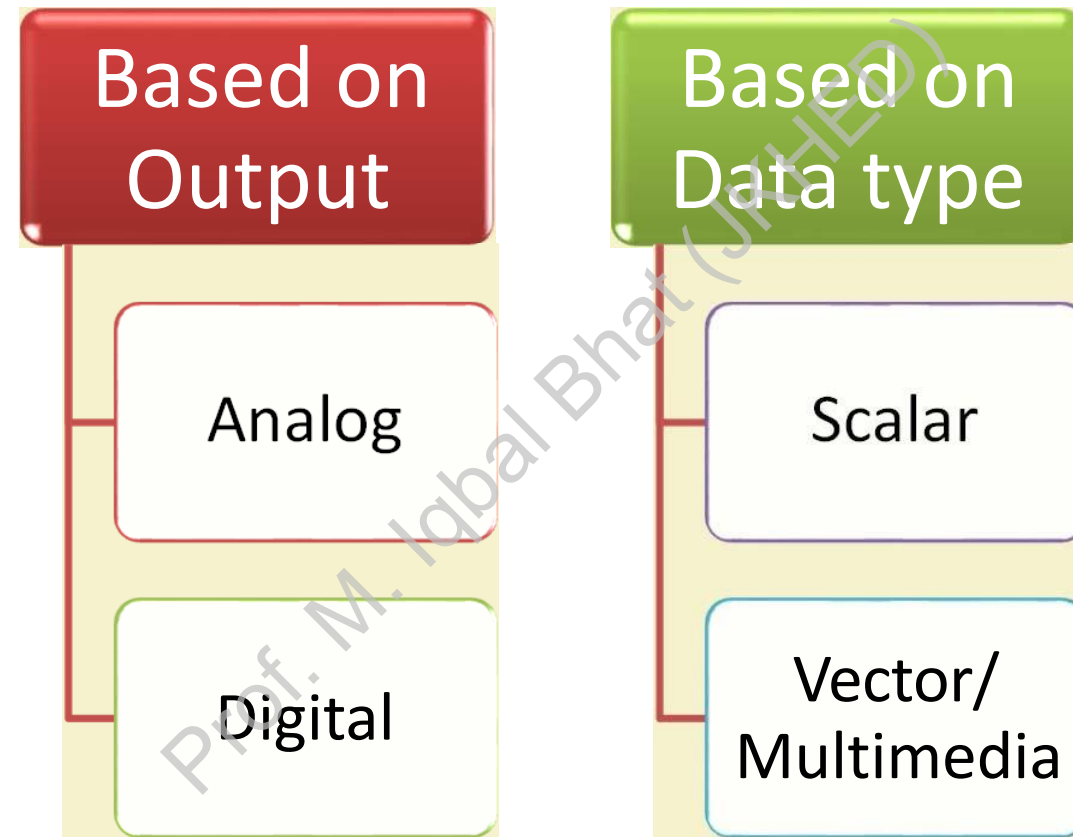
Sensor Features

- ✓ It is only sensitive to the measured property (e.g., A temperature sensor senses the ambient temperature of a room.)
- ✓ It is insensitive to any other property likely to be encountered in its application (e.g., A temperature sensor does not bother about light or pressure while sensing the temperature.)
- ✓ It does not influence the measured property (e.g., measuring the temperature does not reduce or increase the temperature).

Sensor Resolution

- ✓ The resolution of a sensor is the smallest change it can detect in the quantity that it is measuring.
- ✓ The resolution of a sensor with a digital output is usually the smallest resolution the digital output it is capable of processing.
- ✓ The more is the resolution of a sensor, the more accurate is its precision.
- ✓ A sensor's accuracy does not depend upon its resolution.

Sensor Classes



Analog Sensors

- ✓ **Analog Sensors** produce a continuous output signal or voltage which is generally proportional to the quantity being measured.
- ✓ Physical quantities such as Temperature, Speed, Pressure, Displacement, Strain etc. are all analog quantities as they tend to be continuous in nature.
- ✓ For example, the temperature of a liquid can be measured using a thermometer or thermocouple (e.g. in geysers) which continuously responds to temperature changes as the liquid is heated up or cooled down.

Digital Sensors

- ✓ **Digital Sensors** produce discrete digital output signals or voltages that are a digital representation of the quantity being measured.
- ✓ Digital sensors produce a binary output signal in the form of a logic “1” or a logic “0”, (“ON” or “OFF”).
- ✓ Digital signal only produces discrete (non-continuous) values, which may be output as a single “bit” (serial transmission), or by combining the bits to produce a single “byte” output (parallel transmission).

Scalar Sensors

- ✓ **Scalar Sensors** produce output signal or voltage which is generally proportional to the magnitude of the quantity being measured.
- ✓ Physical quantities such as temperature, color, pressure, strain, etc. are all scalar quantities as only their magnitude is sufficient to convey an information.
- ✓ For example, the temperature of a room can be measured using a thermometer or thermocouple, which responds to temperature changes irrespective of the orientation of the sensor or its direction.

Vector Sensors

- ✓ **Vector Sensors** produce output signal or voltage which is generally proportional to the magnitude, direction, as well as the orientation of the quantity being measured.
- ✓ Physical quantities such as sound, image, velocity, acceleration, orientation, etc. are all vector quantities, as only their magnitude is not sufficient to convey the complete information.
- ✓ For example, the acceleration of a body can be measured using an accelerometer, which gives the components of acceleration of the body with respect to the x,y,z coordinate axes.

Sensor Types

Light

- Light Dependent resistor
- Photo-diode

Temperature

- Thermocouple
- Thermistor

Force

- Strain gauge
- Pressure switch

Position

- Potentiometer, Encoders
- Opto-coupler

Speed

- Reflective/ Opto-coupler
- Doppler effect sensor

Sound

- Carbon Microphone
- Piezoelectric Crystal

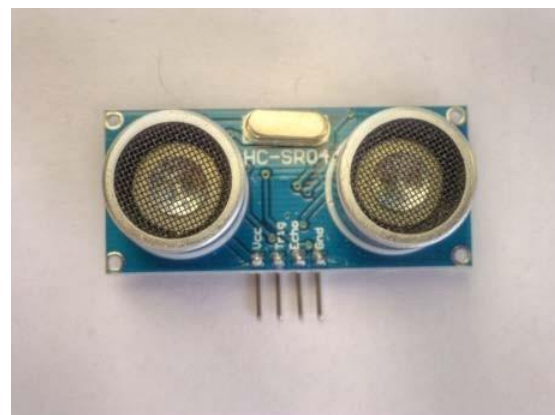
Chemical

- Liquid Chemical sensor
- Gaseous chemical sensor



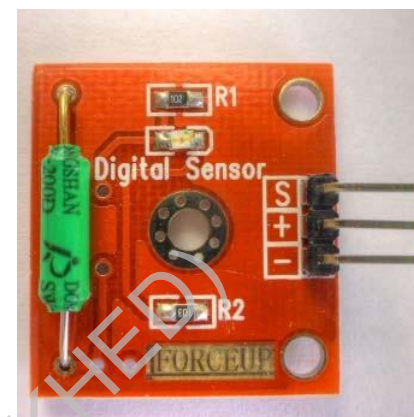
Pressure Sensor

Source: Wikimedia Commons



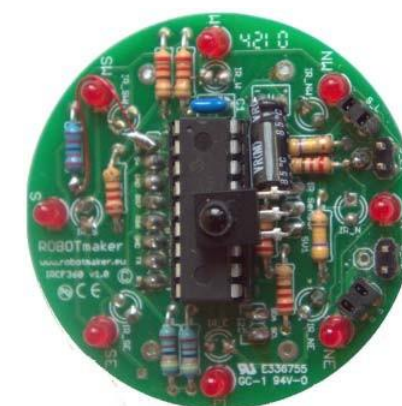
Ultrasonic Distance Sensor

Source: Wikimedia Commons



Tilt Sensor

Source: Wikimedia Commons



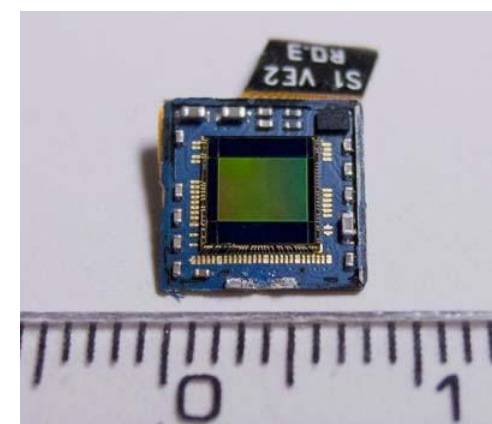
Infrared Motion Sensor

Source: Wikimedia Commons



Analog Temperature Sensor

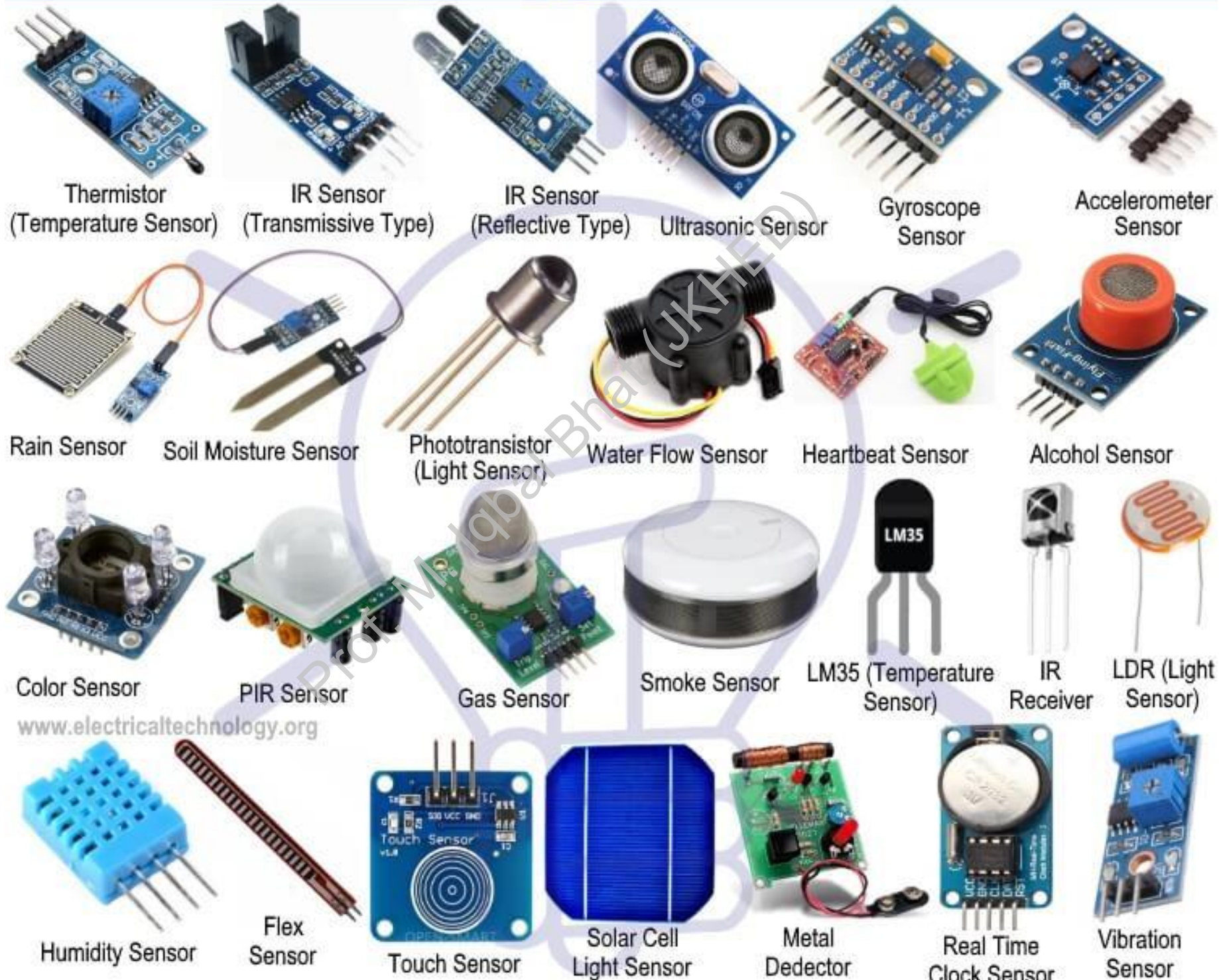
Source: Wikimedia Commons



Camera Sensor

Source: Wikimedia Commons

Different Types of Sensors



Sensorial Deviations

- ✓ Since the range of the output signal is always limited, the output signal will eventually reach a minimum or maximum, when the measured property exceeds the limits. The full scale range of a sensor defines the maximum and minimum values of the measured property.
- ✓ The sensitivity of a sensor under real conditions may differ from the value specified. This is called a **sensitivity error**.
- ✓ If the output signal differs from the correct value by a constant, the sensor has an **offset error** or **bias**.

Reference: <https://en.wikipedia.org/wiki/Sensor>

Non-linearity

- ✓ Nonlinearity is deviation of a sensor's transfer function (TF) from a straight line transfer function.
- ✓ This is defined by the amount the output differs from ideal TF behavior over the full range of the sensor, which is denoted as the percentage of the full range.
- ✓ Most sensors have linear behavior.

Reference: <https://en.wikipedia.org/wiki/Sensor>

- ✓ If the output signal slowly changes independent of the measured property, this is defined as **drift**. Long term drift over months or years is caused by physical changes in the sensor.
- ✓ **Noise** is a random deviation of the signal that varies in time.

Reference: <https://en.wikipedia.org/wiki/Sensor>

Hysteresis Error

- ✓ A hysteresis error causes the sensor output value to vary depending on the sensor's previous input values.
- ✓ If a sensor's output is different depending on whether a specific input value was reached by increasing or decreasing the input, then the sensor has a hysteresis error.
- ✓ The present reading depends on the past input values.
- ✓ Typically in analog sensors, magnetic sensors, heating of metal strips.

Reference: <https://en.wikipedia.org/wiki/Sensor>

Other Errors

- ✓ If the sensor has a digital output, the output is essentially an approximation of the measured property. This error is also called **quantization error**.
- ✓ If the signal is monitored digitally, the sampling frequency can cause a dynamic error, or if the input variable or added noise changes periodically at a frequency proportional to the multiple of the sampling rate, **aliasing errors** may occur.
- ✓ The sensor may to some extent be sensitive to properties other than the property being measured. For example, most sensors are influenced by the temperature of their environment.

Reference: <https://en.wikipedia.org/wiki/Sensor>

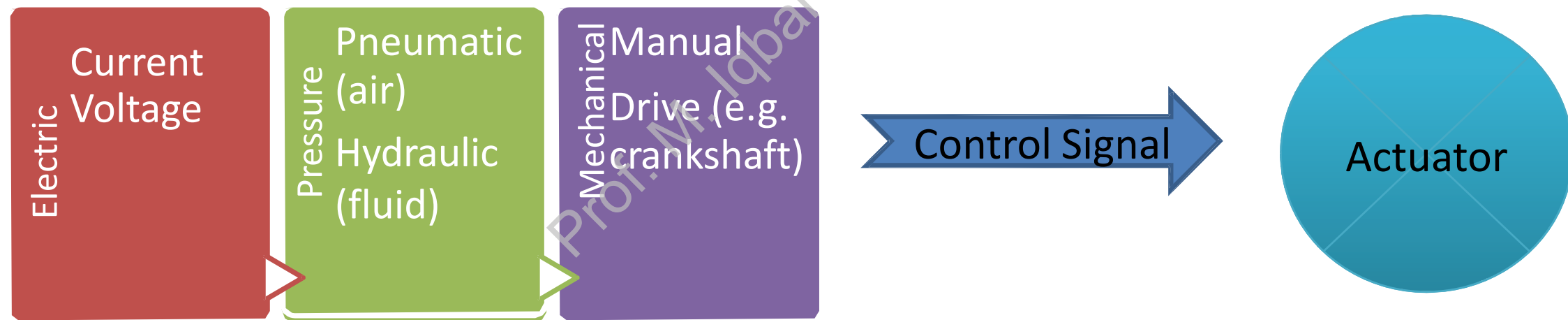
Actuation

Prof. M. Iqbal Bhat (JKHED)

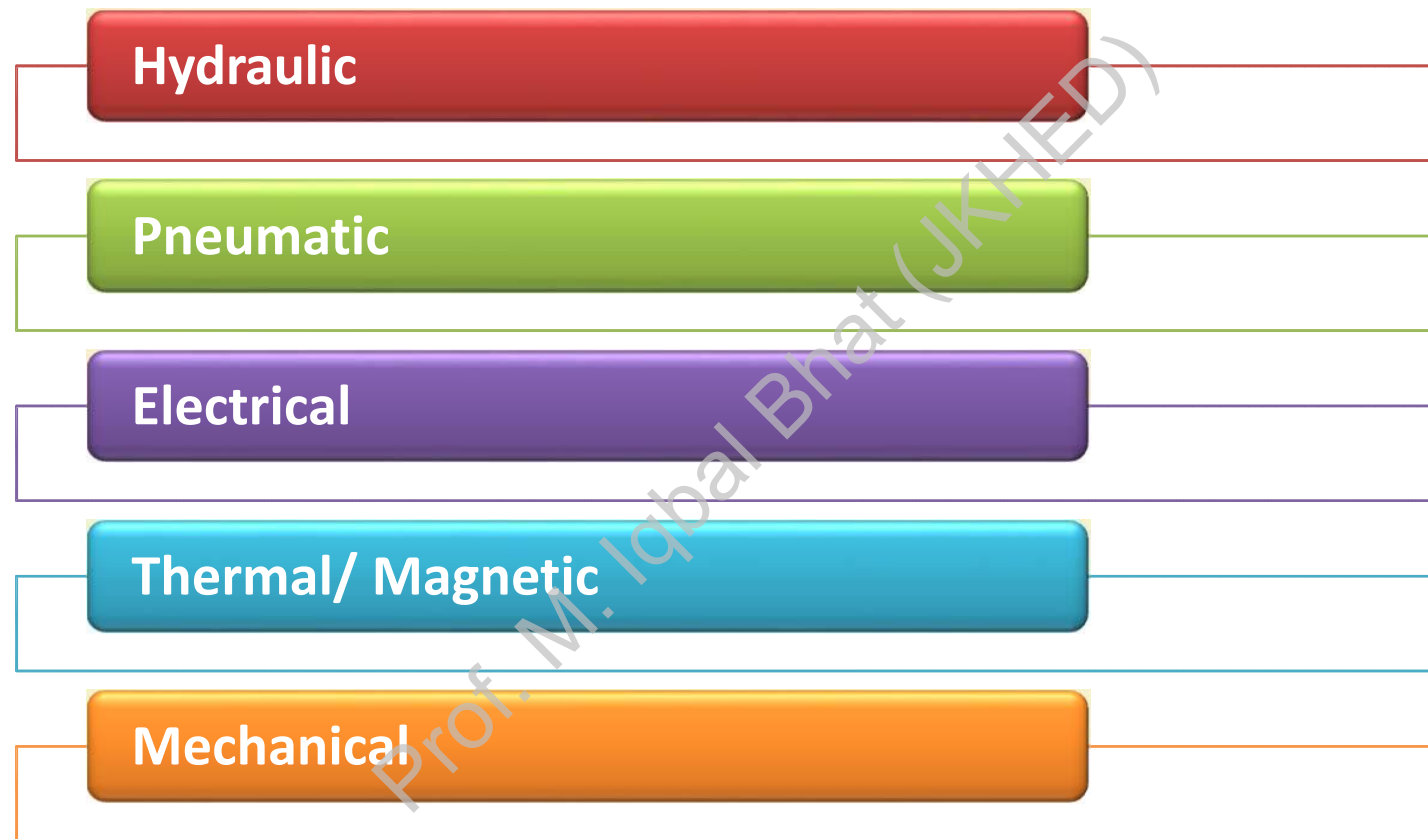
Actuator

- ✓ An actuator is a component of a machine or system that moves or controls the mechanism or the system.
- ✓ An actuator is the mechanism by which a control system acts upon an environment
- ✓ An actuator requires a control signal and a source of energy.

- ✓ Upon receiving a control signal is received, the actuator responds by converting the energy into mechanical motion.
- ✓ The control system can be simple (a fixed mechanical or electronic system), software-based (e.g. a printer driver, robot control system), a human, or any other input.



Actuator Types



Hydraulic Actuators

- ✓ A hydraulic actuator consists of a cylinder or fluid motor that uses hydraulic power to facilitate mechanical operation.
- ✓ The mechanical motion is converted to linear, rotary or oscillatory motion.
- ✓ Since liquids are nearly impossible to compress, a hydraulic actuator exerts considerable force.
- ✓ The actuator's limited acceleration restricts its usage.

Reference: <https://en.wikipedia.org/wiki/Actuator>

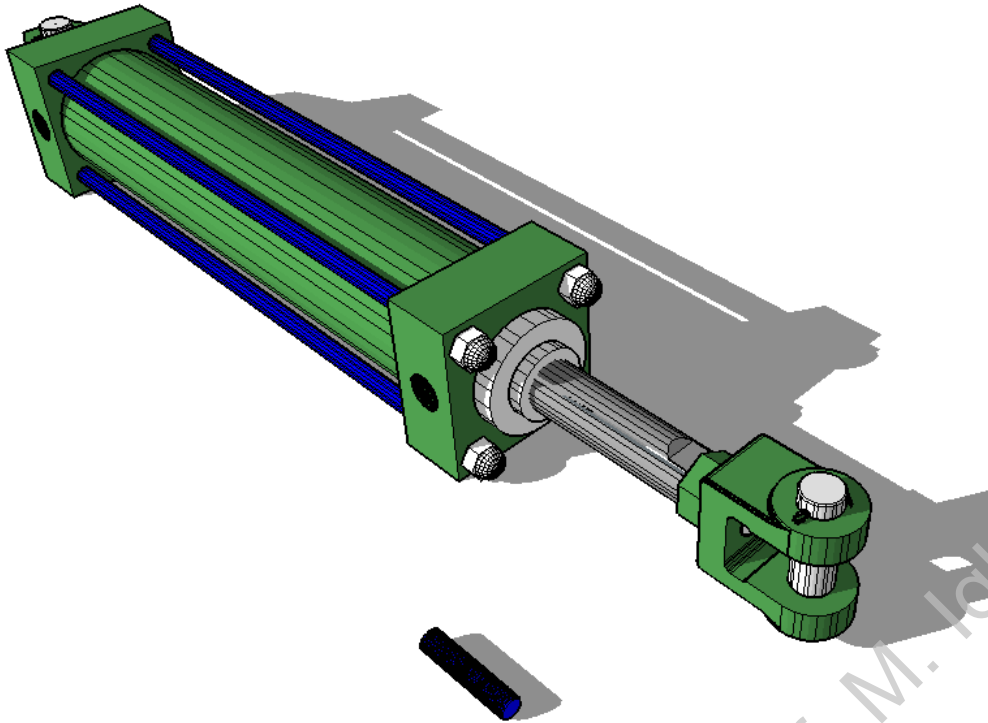


Fig: An oil based hydraulic actuator

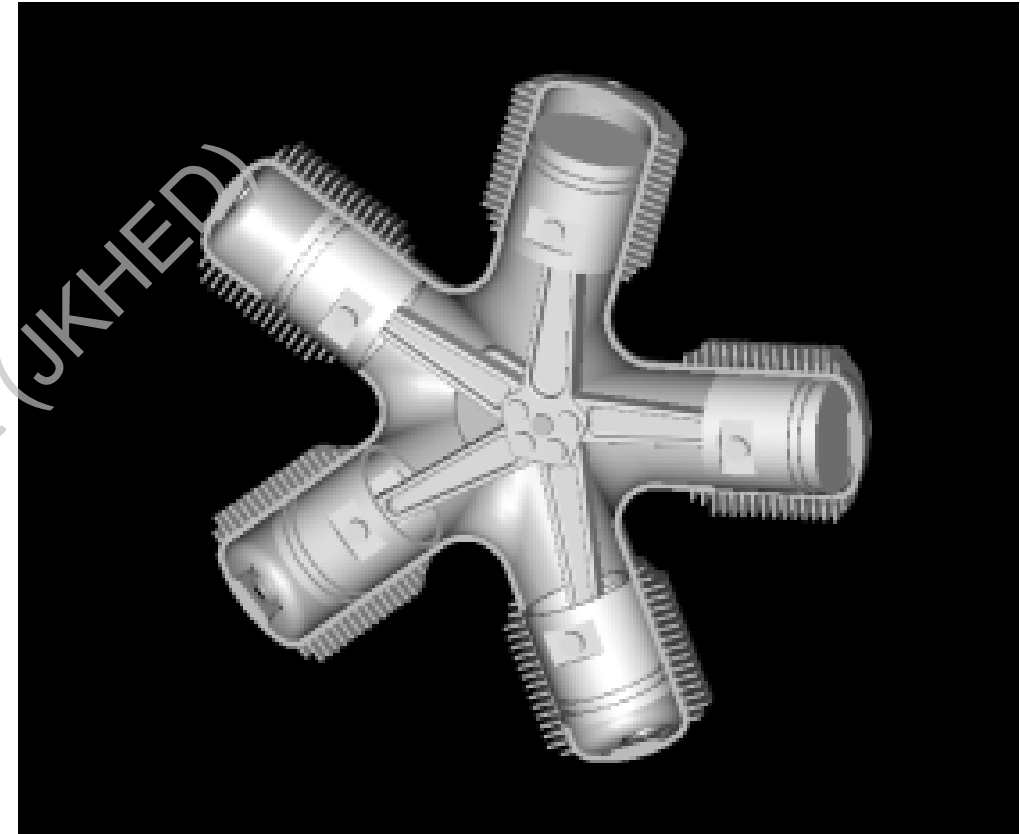


Fig: A radial engine acts as a hydraulic actuator

Source: Wikimedia Commons

File: Radial_engine.gif

Pneumatic Actuators

- ✓ A pneumatic actuator converts energy formed by vacuum or compressed air at high pressure into either linear or rotary motion.
- ✓ Pneumatic rack and pinion actuators are used for valve controls of water pipes.
- ✓ Pneumatic energy quickly responds to starting and stopping signals.
- ✓ The power source does not need to be stored in reserve for operation.

Reference: <https://en.wikipedia.org/wiki/Actuator>

- ✓ Pneumatic actuators enable large forces to be produced from relatively small pressure changes (e.g., Pneumatic brakes can are very responsive to small changes in pressure applied by the driver).
- ✓ It is responsible for converting pressure into force.

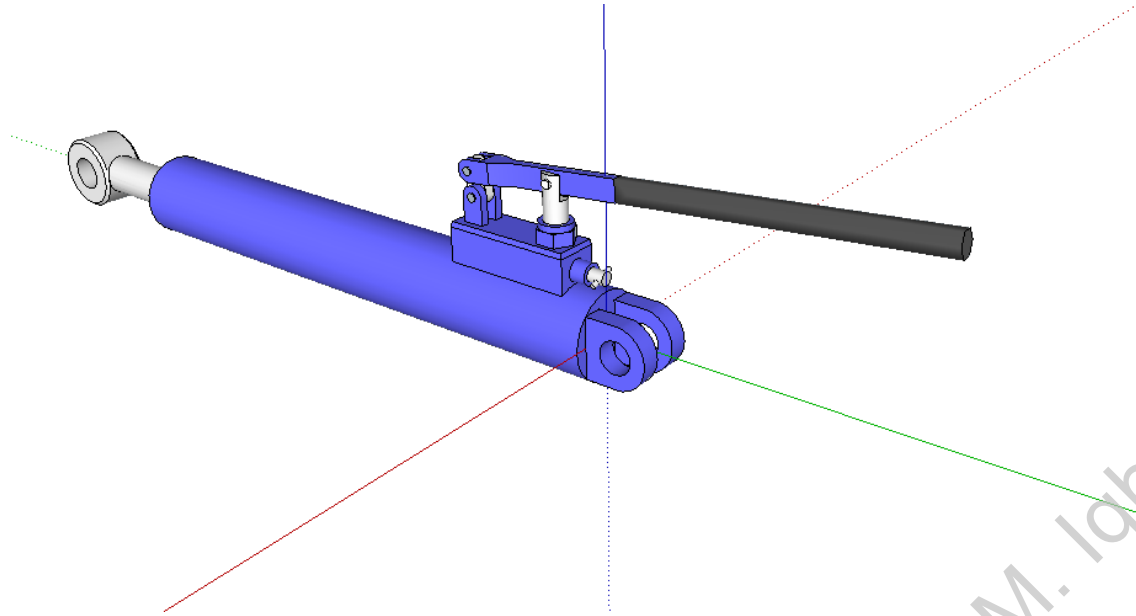


Fig: A manual linear pneumatic actuator

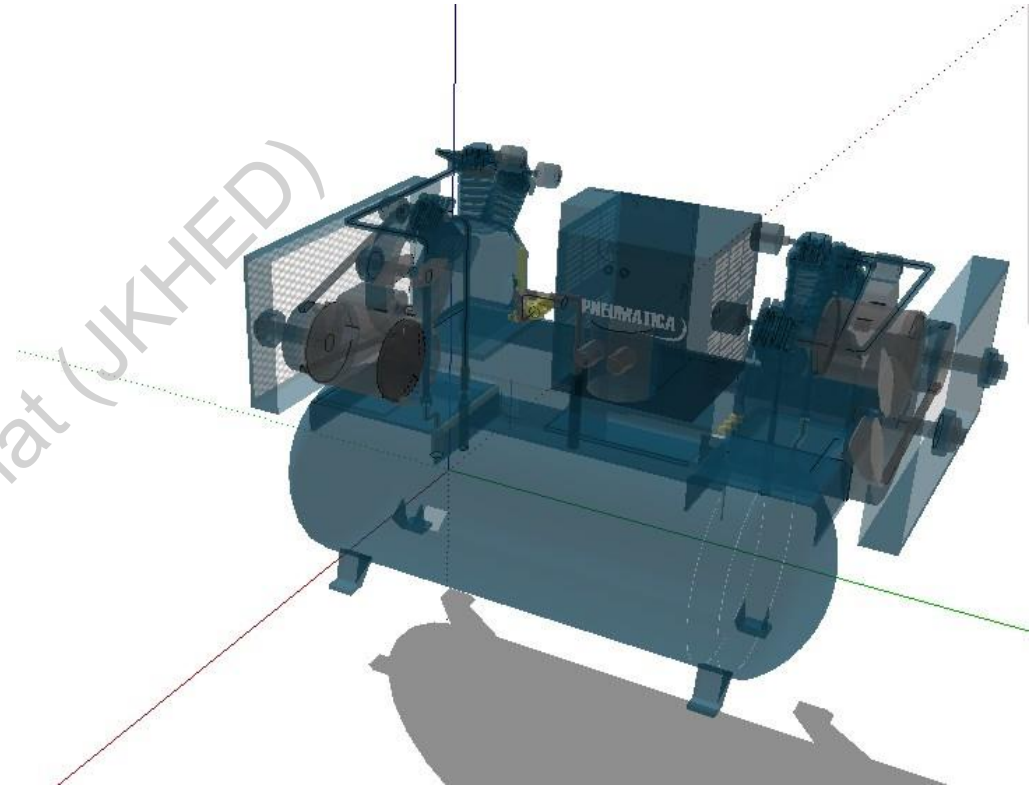


Fig: An air pump acts as a pneumatic actuator

Electric Actuators

- ✓ An electric actuator is generally powered by a motor that converts electrical energy into mechanical torque.
- ✓ The electrical energy is used to actuate equipment such as solenoid valves which control the flow of water in pipes in response to electrical signals.
- ✓ Considered as one of the cheapest, cleanest and speedy actuator types available.

Reference: <https://en.wikipedia.org/wiki/Actuator>

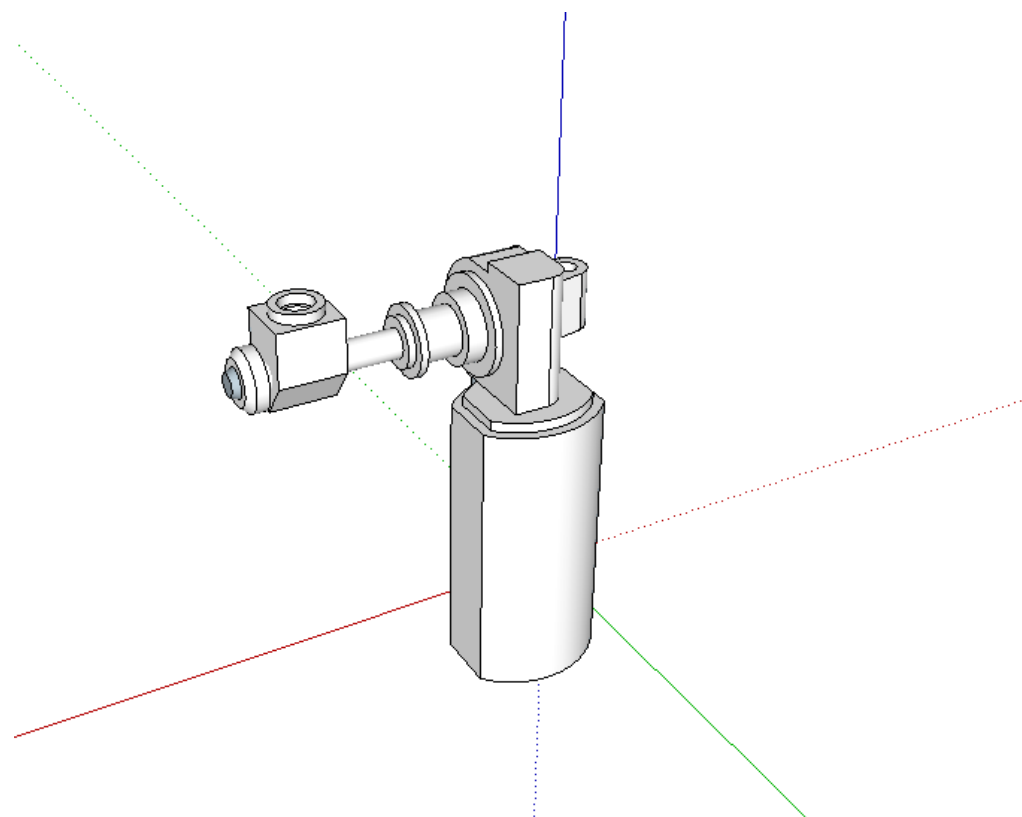


Fig: A motor drive-based rotary actuator

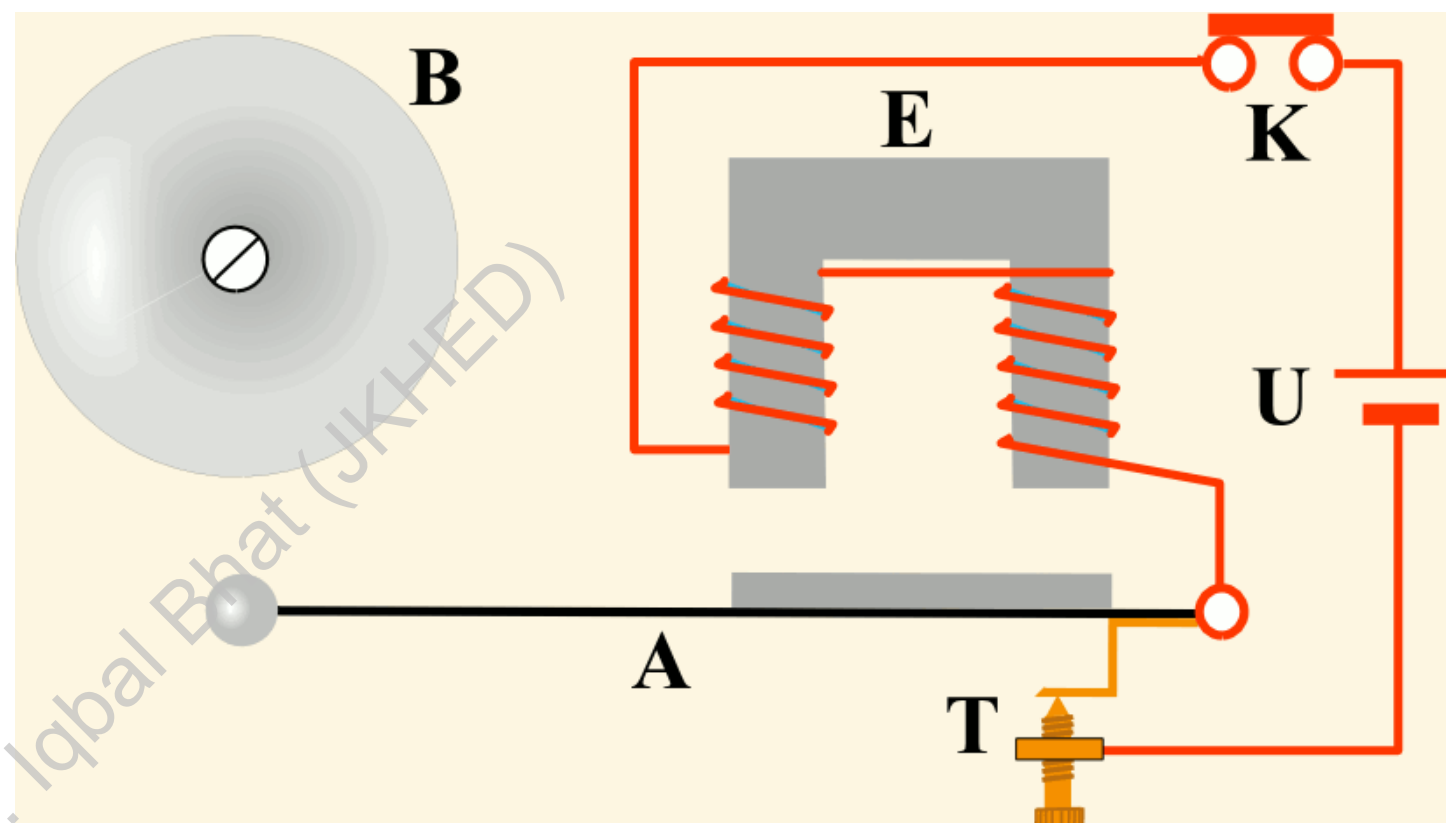


Fig: A solenoid based electric bell ringing mechanism

Source: Wikimedia Commons

File: Electric_Bell_animation.gif

Thermal or Magnetic Actuators

- ✓ These can be actuated by applying thermal or magnetic energy.
- ✓ They tend to be compact, lightweight, economical and with high power density.
- ✓ These actuators use shape memory materials (SMMs), such as shape memory alloys (SMAs) or magnetic shape-memory alloys (MSMAs).
- ✓ Some popular manufacturers of these devices are *Finnish Modti Inc.* and *American Dynalloy*.

Reference: <https://en.wikipedia.org/wiki/Actuator>

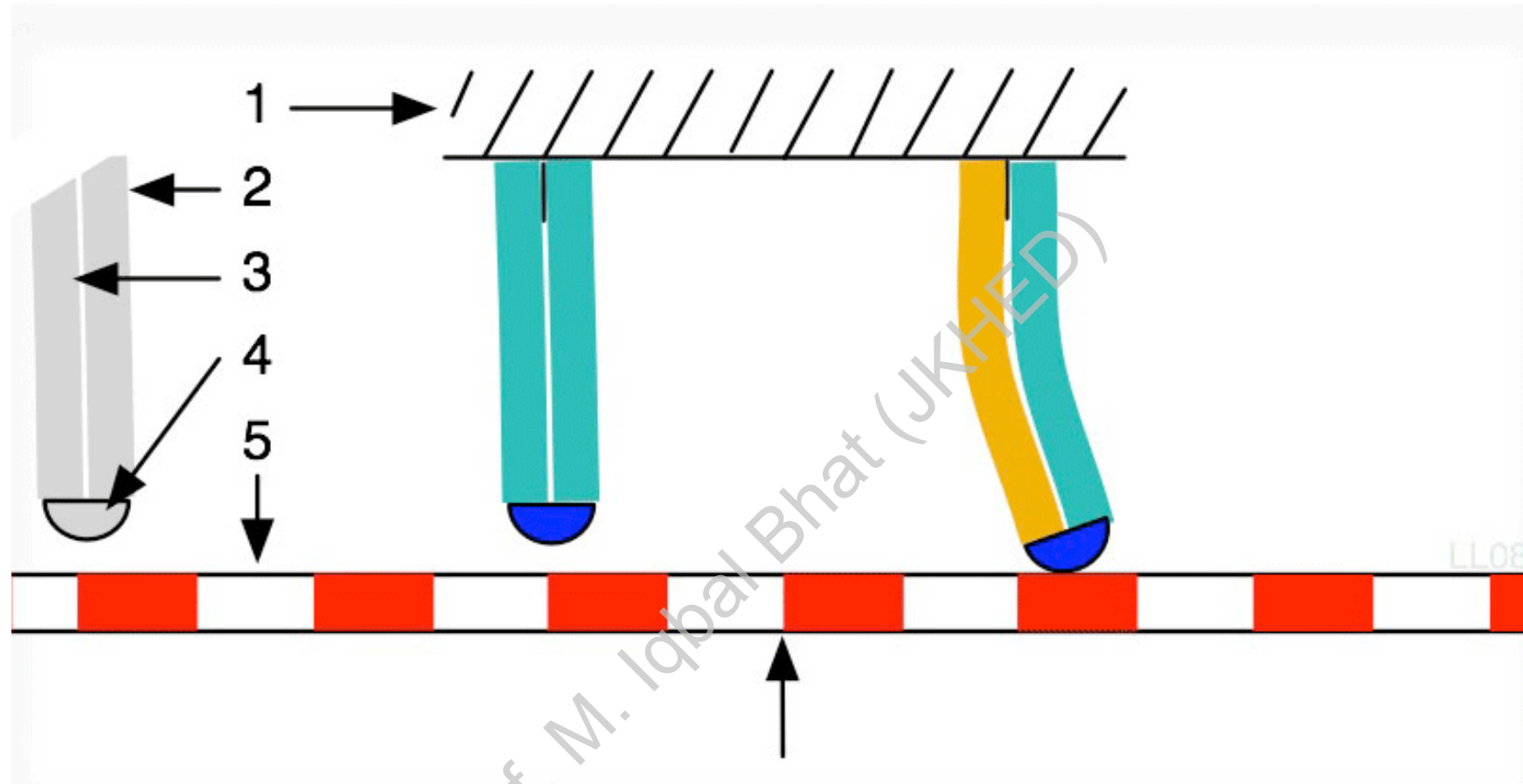


Fig: A piezo motor using SMA

Source: Wikimedia Commons

File: Piezomotor type bimorph.gif

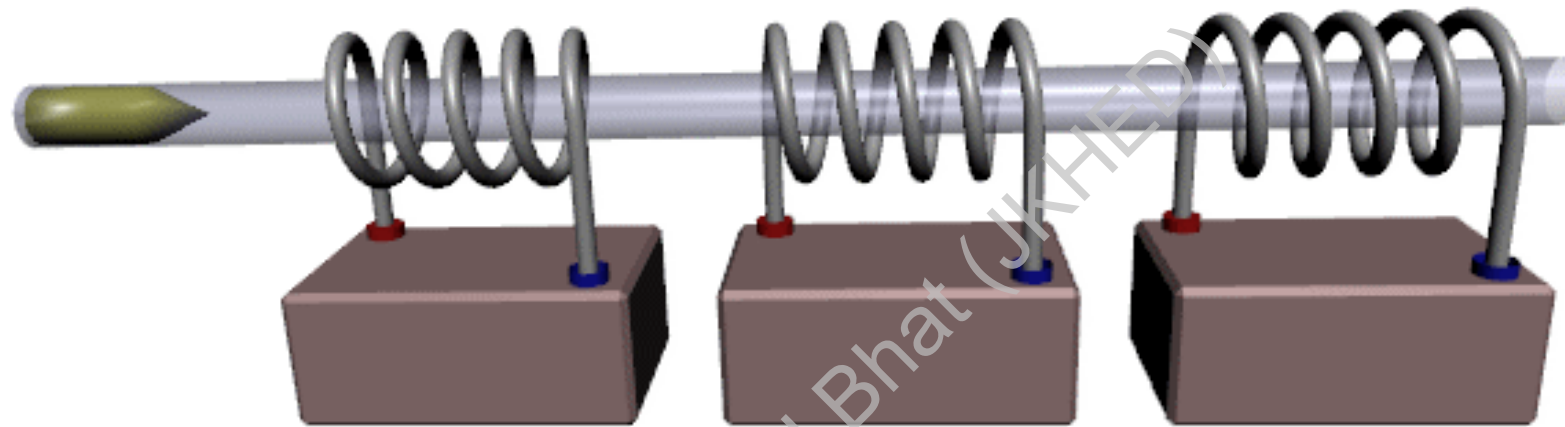


Fig: A coil gun works on the principle of magnetic actuation

Source: Wikimedia Commons

File: Coilgun animation.gif

Mechanical Actuators

- ✓ A mechanical actuator converts rotary motion into linear motion to execute some movement.
- ✓ It involves gears, rails, pulleys, chains and other devices to operate.
- ✓ Example: rack and pinion.



Fig: A rack and pinion mechanism

Source: Wikimedia Commons

File: Rack and pinion.png

Reference: <https://en.wikipedia.org/wiki/Actuator>

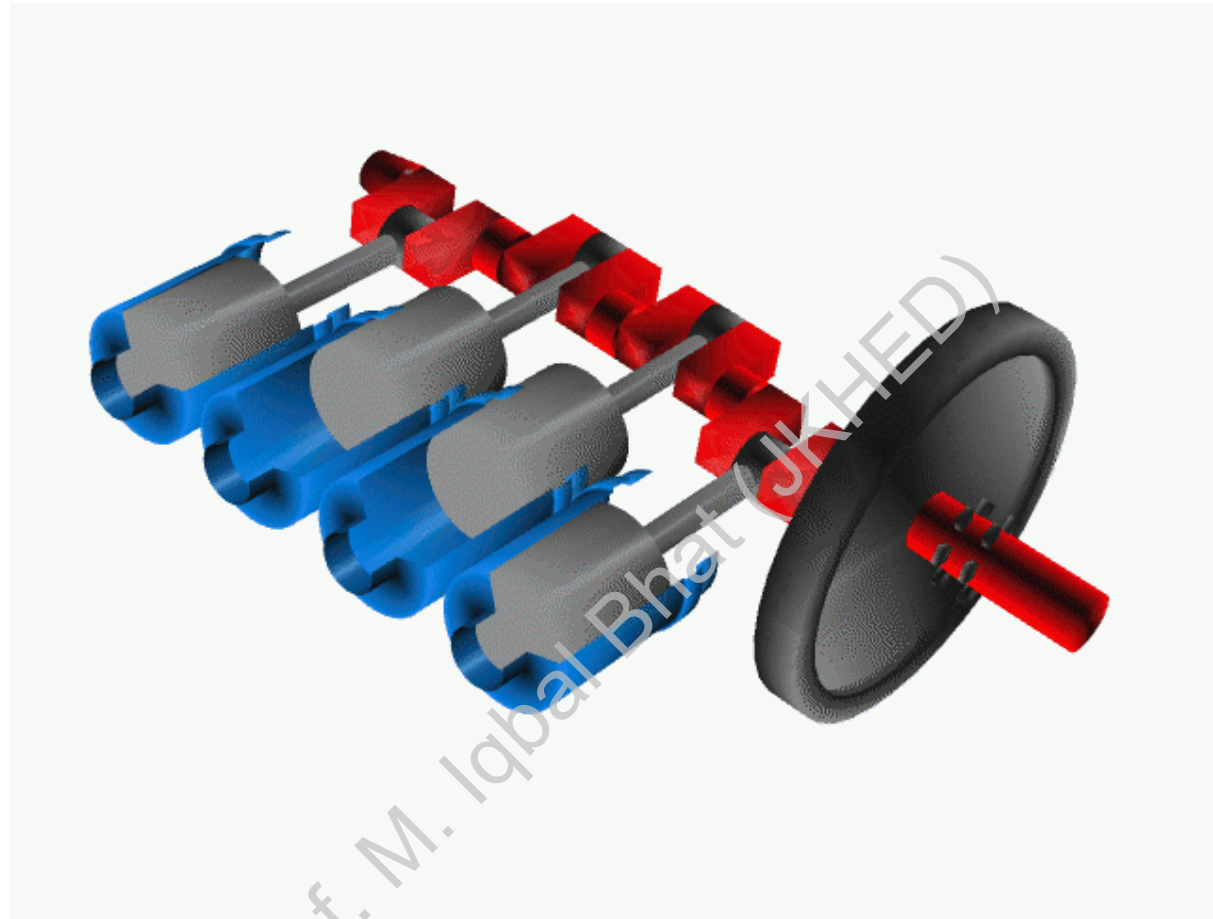


Fig: A crank shaft acting as a mechanical actuator

Source: Wikimedia Commons

File: Cshaft.gif

Soft Actuators

- ✓ Soft actuators (e.g. polymer based) are designed to handle fragile objects like fruit harvesting in agriculture or manipulating the internal organs in biomedicine.
- ✓ They typically address challenging tasks in robotics.
- ✓ Soft actuators produce flexible motion due to the integration of microscopic changes at the molecular level into a macroscopic deformation of the actuator materials.

Reference: <https://en.wikipedia.org/wiki/Actuator>

Shape Memory Polymers

- ✓ Shape memory polymer (SMP) actuators function similar to our muscles, even providing a response to a range of stimuli such as light, electrical, magnetic, heat, pH, and moisture changes.
- ✓ SMP exhibits surprising features such a low density, high strain recovery, biocompatibility, and biodegradability.

Reference: <https://en.wikipedia.org/wiki/Actuator>

Light Activated Polymers

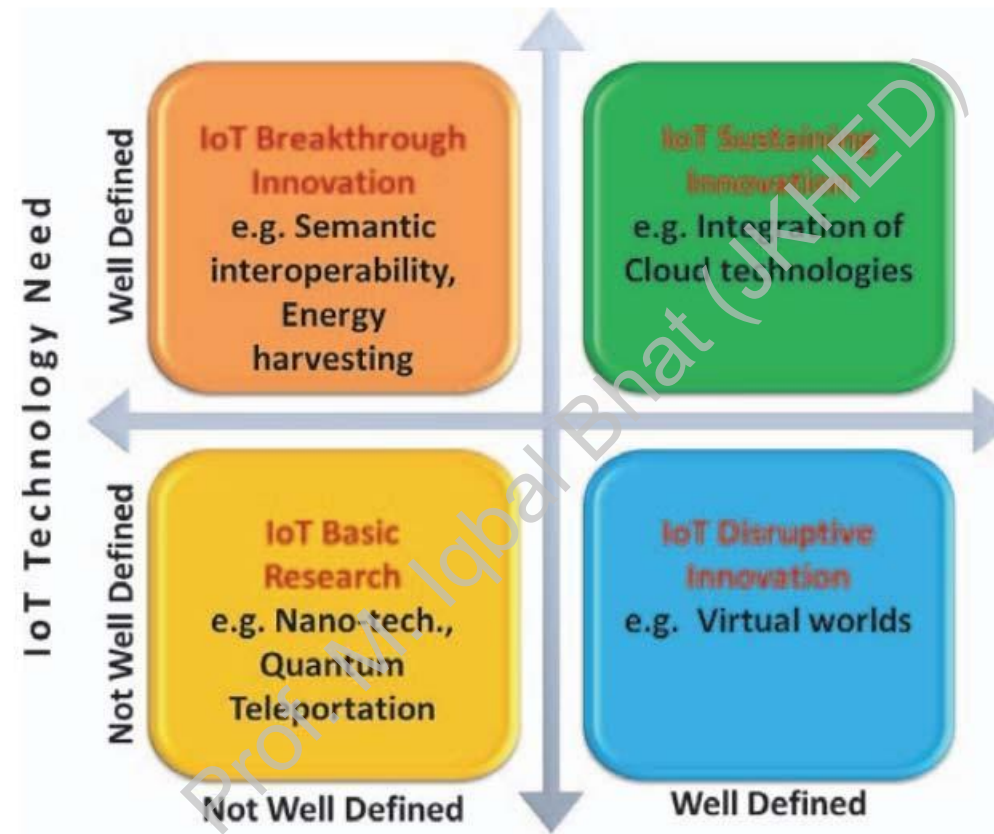
- ✓ Photopolymer/light activated polymers (LAP) are a special type of SMP that are activated by light stimuli.
- ✓ The LAP actuators have instant response.
- ✓ They can be controlled remotely without any physical contact, only using the variation of light frequency or intensity.

Reference: <https://en.wikipedia.org/wiki/Actuator>

IoT Networking |

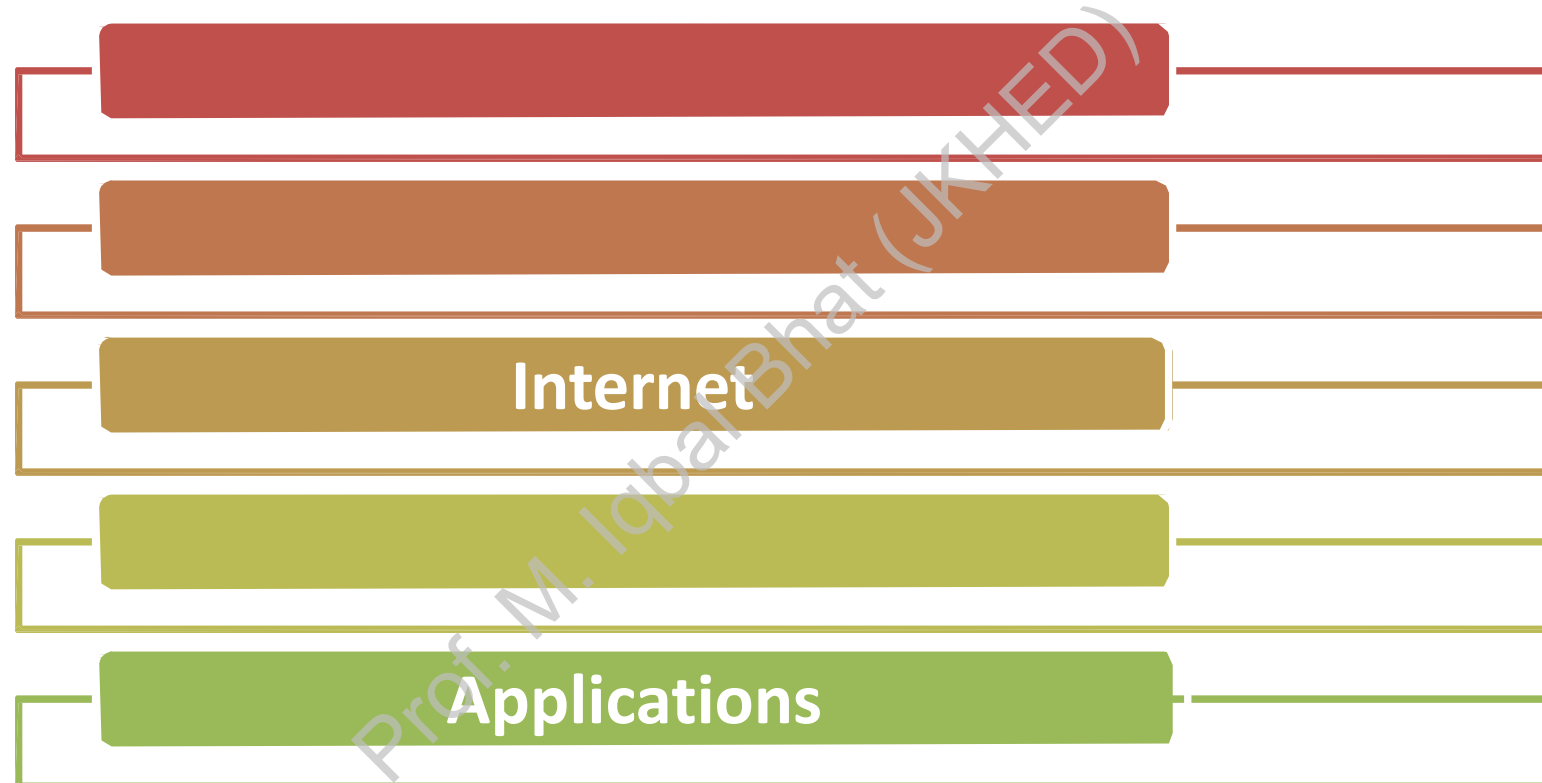
Prof. M. Iqbal Bhat (JKHED)

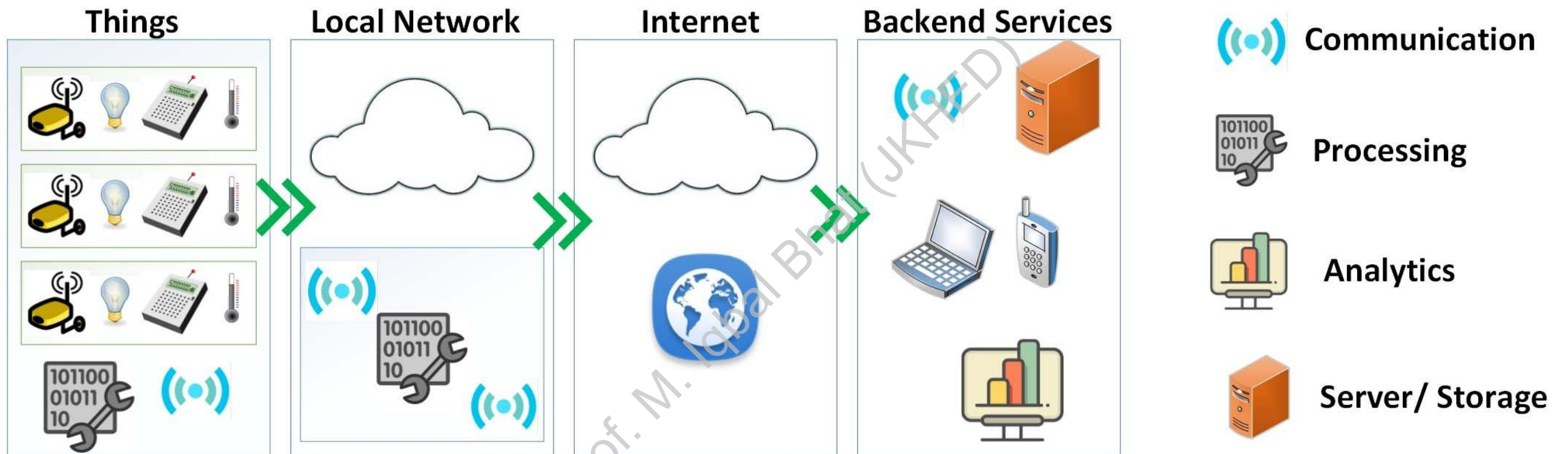
Convergence of Domains



Source: O. Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013

IoT Components



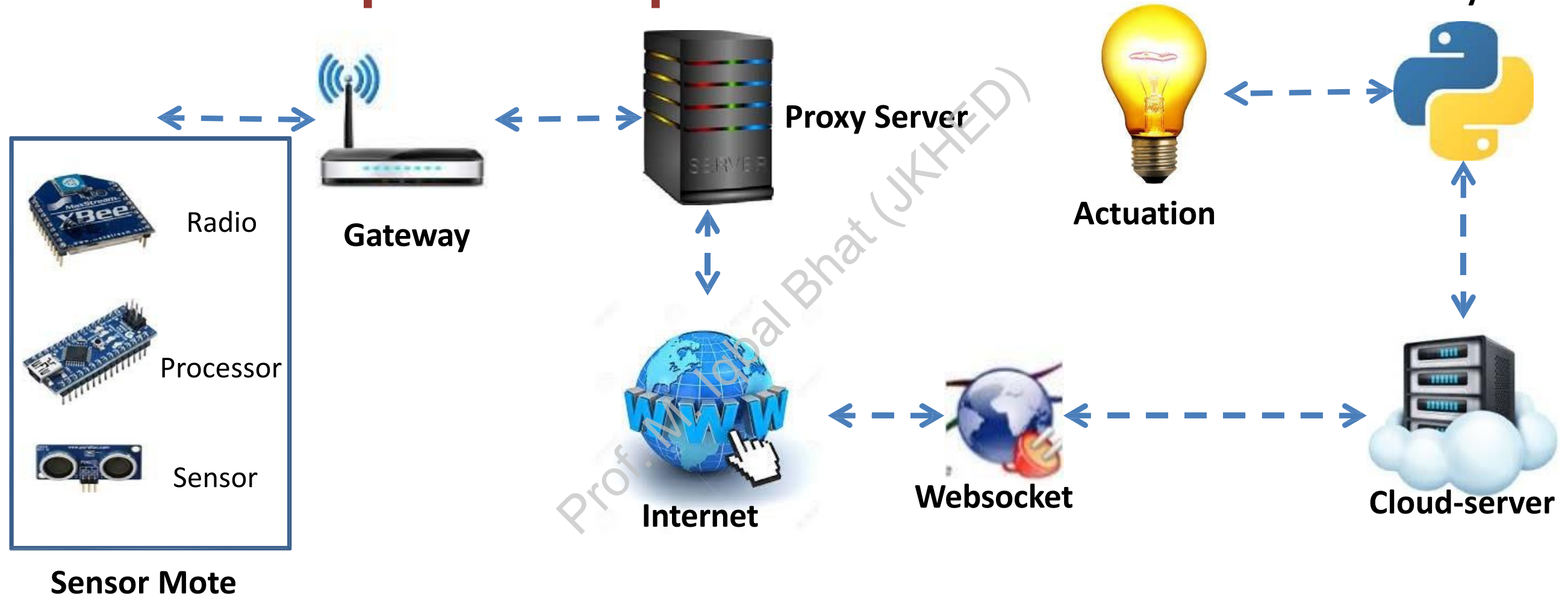


Functional Components of IoT

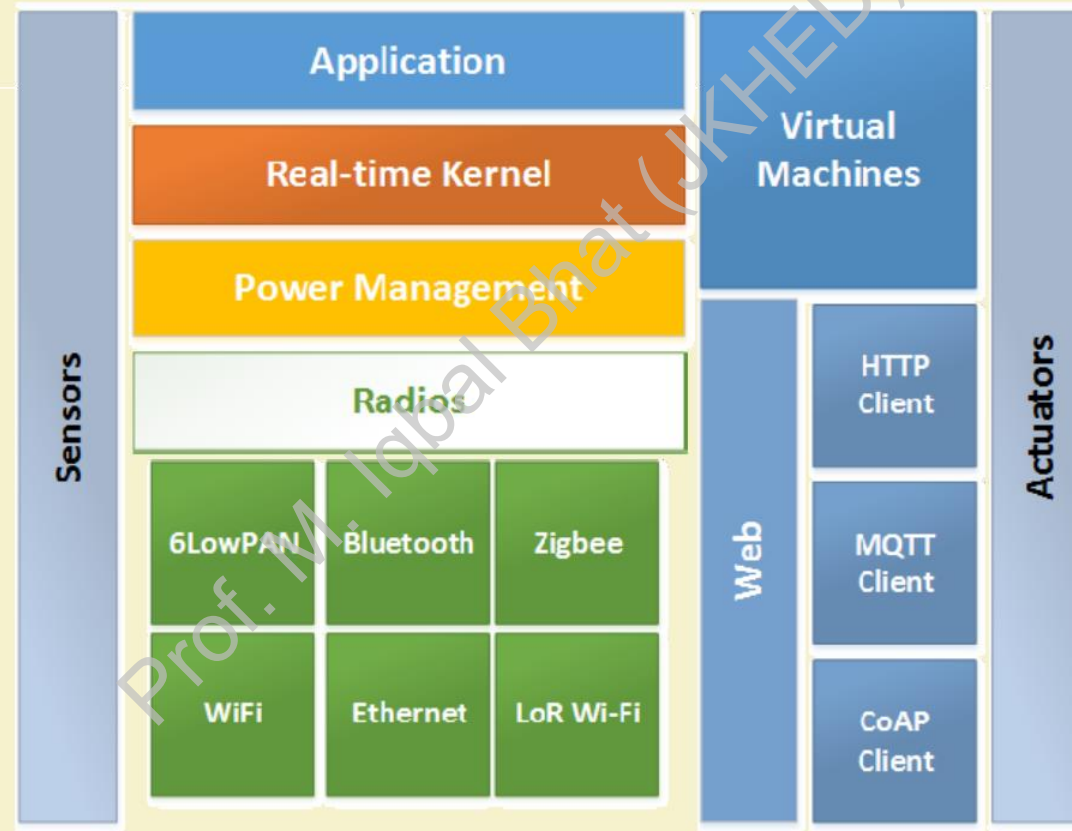
- ✓ Component for interaction and communication with other IoT devices
- ✓ Component for processing and analysis of operations
- ✓ Component for Internet interaction
- ✓ Components for handling Web services of applications
- ✓ Component to integrate application services
- ✓ User interface to access IoT

Source: O Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013

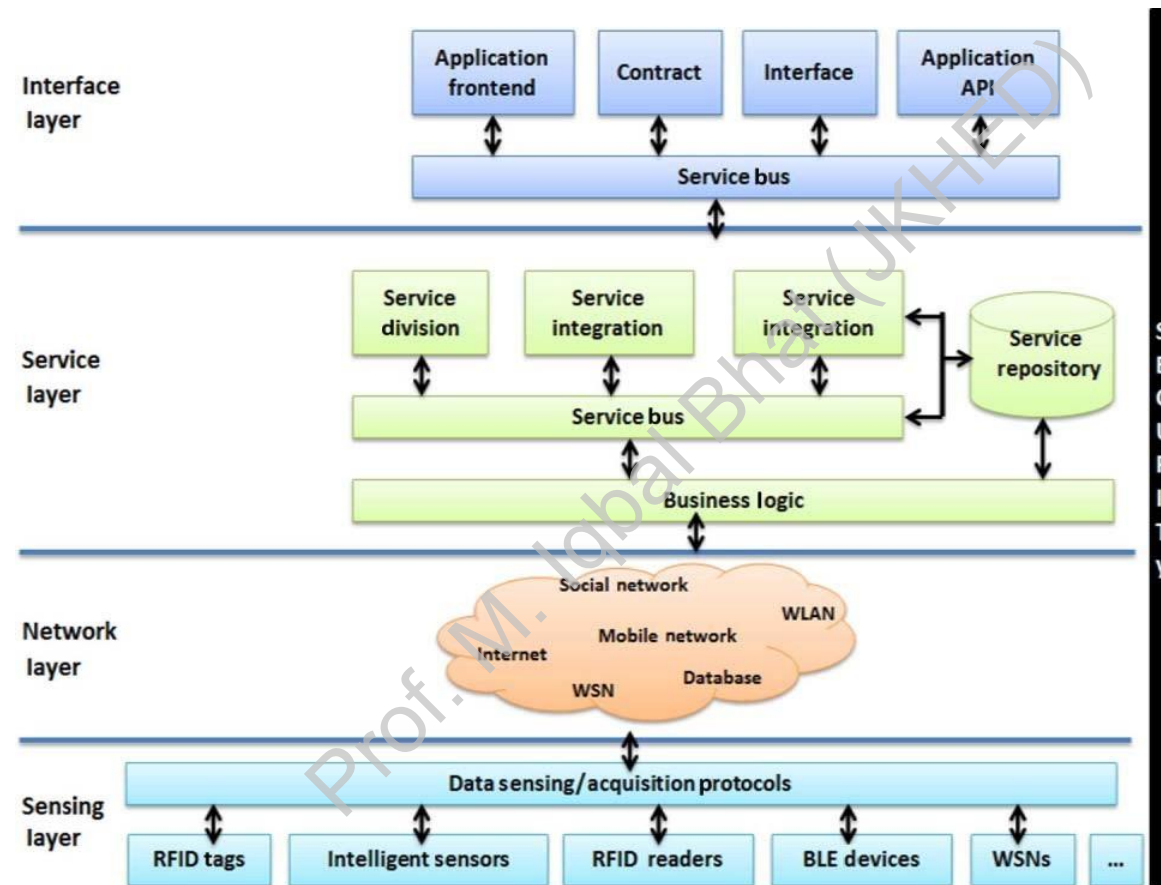
An Example IoT Implementation



IoT Interdependencies



IoT Service Oriented Architecture



Source: Li Da Xu, Wu He, and Shancang Li, "Internet of Things in Industries: A Survey ", IEEE Transactions on Industrial Informatics, Vol. 10, No. 4, Nov. 2014.

IoT Categories

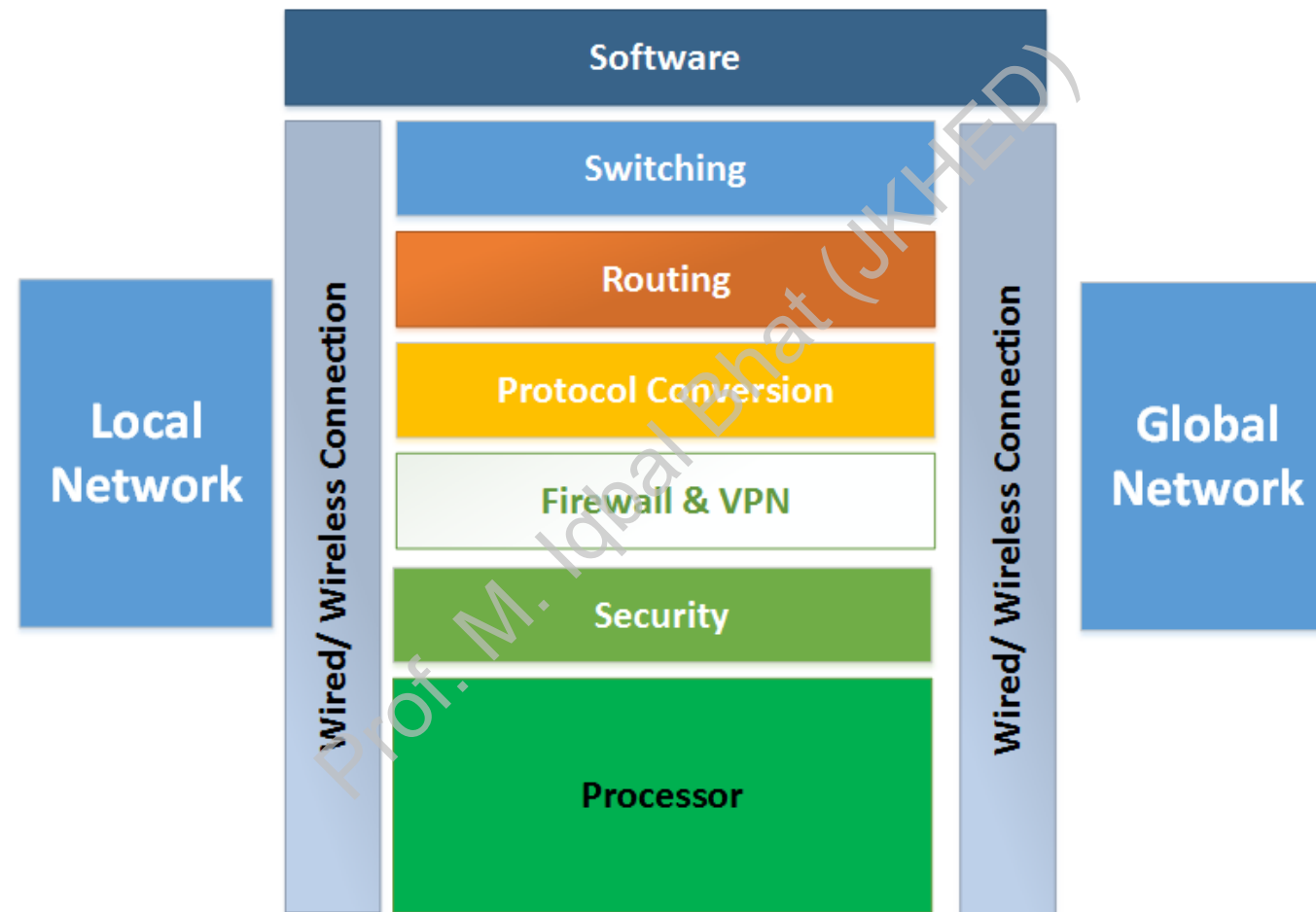
✓ Industrial IoT

- IoT device connects to an IP network and the global Internet.
- Communication between the nodes done using regular as well as industry specific technologies.

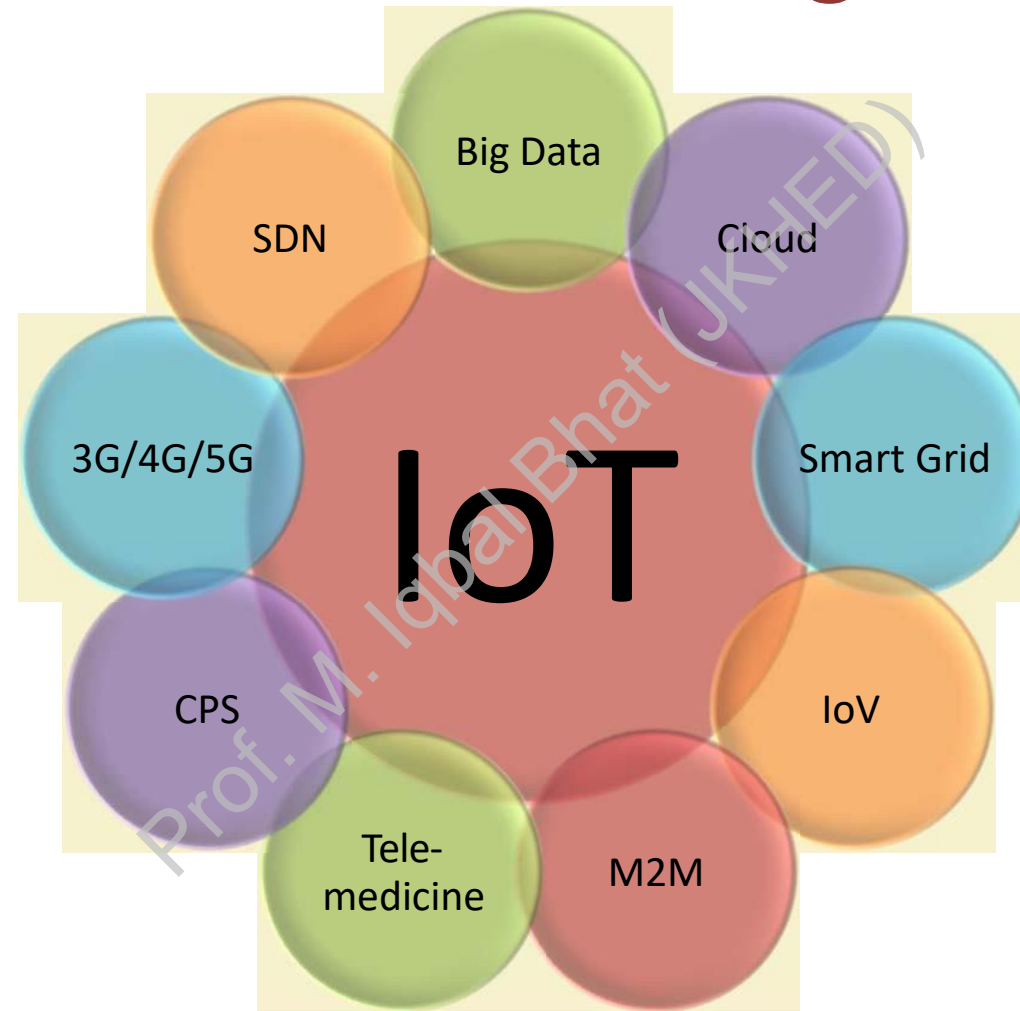
✓ Consumer IoT

- IoT device communicates within the locally networked devices.
- Local communication is done mainly via Bluetooth, Zigbee or WiFi.
- Generally limited to local communication by a Gateway

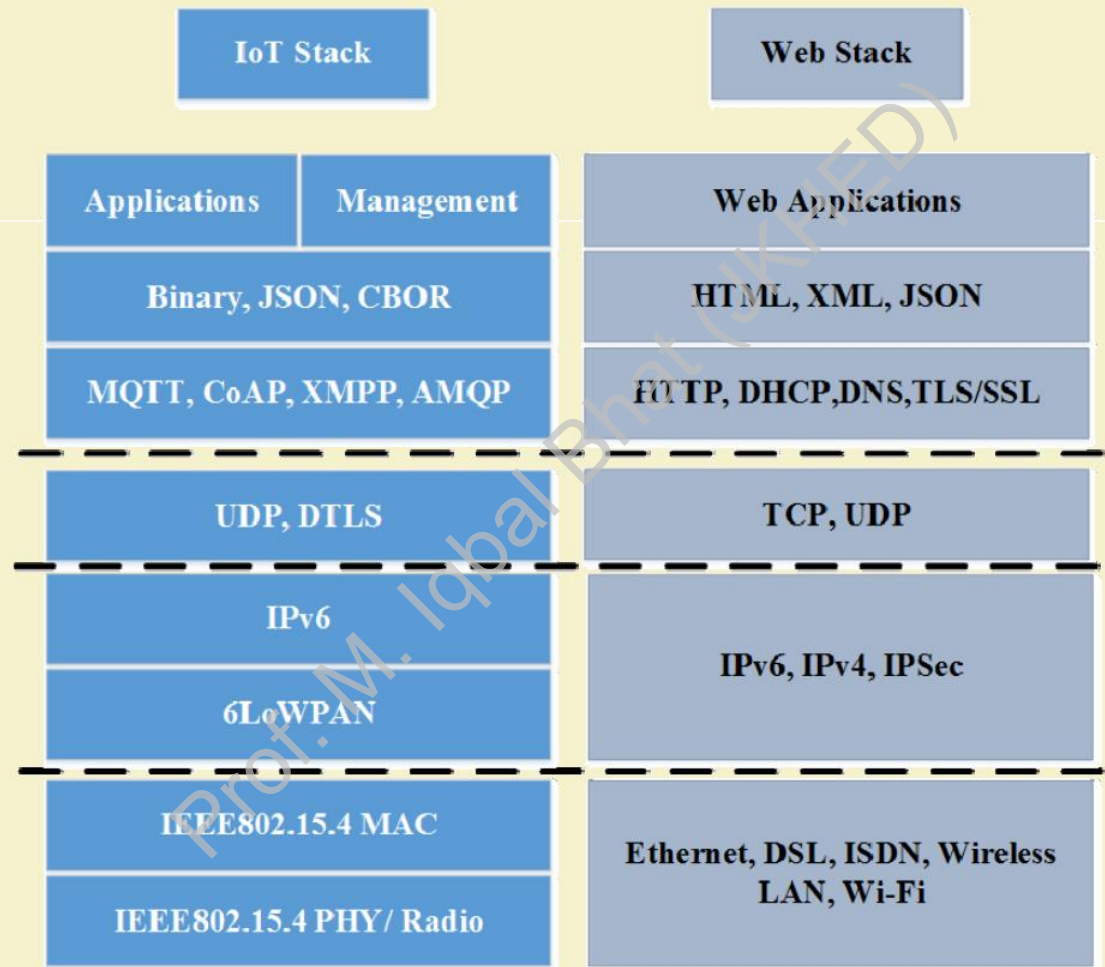
IoT Gateways



IoT and Associated Technologies



Technical Deviations from Regular Web



Key Technologies for IoT



Source: O Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013

IoT Challenges

- ✓ Security
- ✓ Scalability
- ✓ Energy efficiency
- ✓ Bandwidth management
- ✓ Modeling and Analysis
- ✓ Interfacing
- ✓ Interoperability
- ✓ Data storage
- ✓ Data Analytics
- ✓ Complexity management (e.g., SDN)

Considerations

- ✓ Communication between the IoT device(s) and the outside world dictates the network architecture.
- ✓ Choice of communication technology dictates the IoT device hardware requirements and costs.
- ✓ Due to the presence of numerous applications of IoT enabled devices, a single networking paradigm not sufficient to address all the needs of the consumer or the IoT device.

Complexity of Networks

- ✓ Growth of networks
- ✓ Interference among devices
- ✓ Network management
- ✓ Heterogeneity in networks
- ✓ Protocol standardization within networks

Source: O Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013

Wireless Networks

- Traffic and load management
- Variations in wireless networks – Wireless Body Area Networks and other Personal Area Networks
- Interoperability
- Network management
- Overlay networks

Source: O. Vermesan, P. Friess, “Internet of Things – Converging Technologies for Smart Environments and Integrated Ecosystems”, River Publishers, Series in Communications, 2013

Scalability

- Flexibility within Internet
- IoT integration
- Large scale deployment
- Real-time connectivity of billions of devices

Thank You!!

Prof. M. Iqbal Bhat (JKHED)